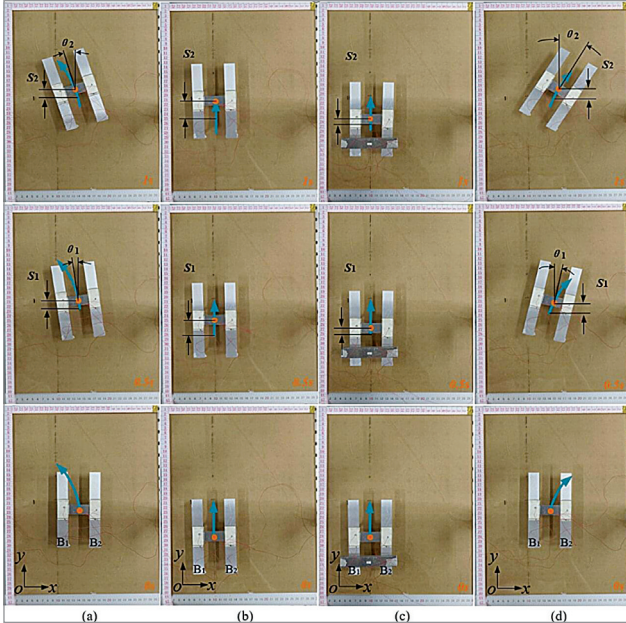


Bionic Robot Inspired By Cheetahs

Researchers at Northeastern University, China, have developed an H-shaped bionic piezoelectric robot (H-BPR) mimicking a cheetah's gait. Using piezoelectric materials,



Experimental motion trajectory of the robot (a) Load 0gm, $VB1=0V$, $VB2=400V$; (b) Load 0gm, $VB1=400V$, $VB2=400V$; (c) Load 55gm, $VB1=200V$, $VB2=200V$; (d) Load 0gm, $VB1=400V$, $VB2=0V$ (Credit: *Journal of Bionic Engineering*, 2025)

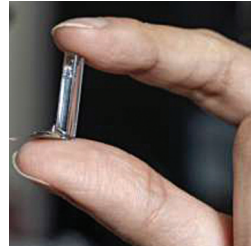
the robot moves via bending vibrations of three beams connecting its four legs. Unlike traditional piezoelectric robots, this design simplifies manufacturing and allows movement adjustments via voltage control. The prototype reached 66.79mm/s at 320V and carried 55gm. Unequal drive legs improved climbing performance. Future enhancements may include sensors, extreme-condition resistance, and industrial applications like search-and-rescue. Real-world tests confirmed its efficiency and adaptability.

Smart Wearables With Failure Control

Researchers at Rice University have enhanced sheet-based fluidic devices, vital for soft robotics and medical wearables, by introducing 'programmed failure.' Their method improves reliability, prevents damage, and simplifies control. Using heat-sealed flexible sheets, they identified three failure patterns based on bond geometry and material selection. This led to a 'fluidic fuse,' a replaceable component that breaks at a set pressure, preventing system failure. The innovation enables complex tasks with a single input, benefiting robotics, medical wearables, and automation by reducing reliance on intricate electronics.

Engineers Defy Physics For Clean Energy

A team at the University of Colorado Boulder, led by Assistant Professor Longji Cui, has revolutionised energy generation



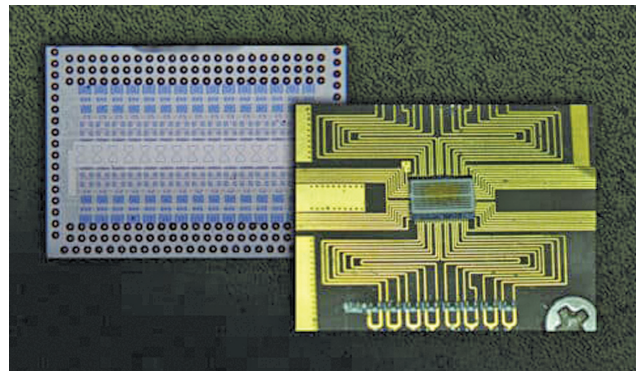
One of the group's TPV cells used for power generation (Credit: CU Boulder College of Engineering and Applied Science)

tion by surpassing Planck's Law in thermal radiation conversion. Collaborating with NREL and the University of Wisconsin-Madison, they developed a thermophotovoltaic (TPV) device that doubles power output. Instead of a vacuum gap, their design uses an infrared-transparent glass spacer, enhancing efficiency at 1000°C. Future advancements with amorphous silicon could boost power density 20-fold, benefiting heavy

industries and reducing carbon emissions. With a patent pending, commercialisation is on the horizon.

Chip-Based Terahertz Wave Generator

MIT researchers have developed a terahertz amplifier-multiplier system that enhances radiating power without bulky silicon lenses, enabling compact integration for applications like security scanners. They applied electromechanical matching theory, balancing silicon, and air's dielectric



By affixing a thin, patterned sheet of material to the back of the chip, highlighted in the centre and shown in the left-side micrograph, the researchers produced a more efficient, yet scalable, chip-based terahertz wave generator (Credit: Courtesy of the researchers; MIT News)

constants using a thin, patterned sheet to minimise signal reflection. A laser-cut, low-cost substrate fine-tunes properties, while Intel's high-power transistors boost efficiency. Their chip generates 11.1dBm peak power, outperforming existing methods. To scale production, they're developing a phased array of CMOS terahertz sources for steerable, high-powered beams in cost-effective, compact devices.